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2014.0 RANGE ROVER (LG), 303-13

EVAPORATIVE EMISSIONS - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

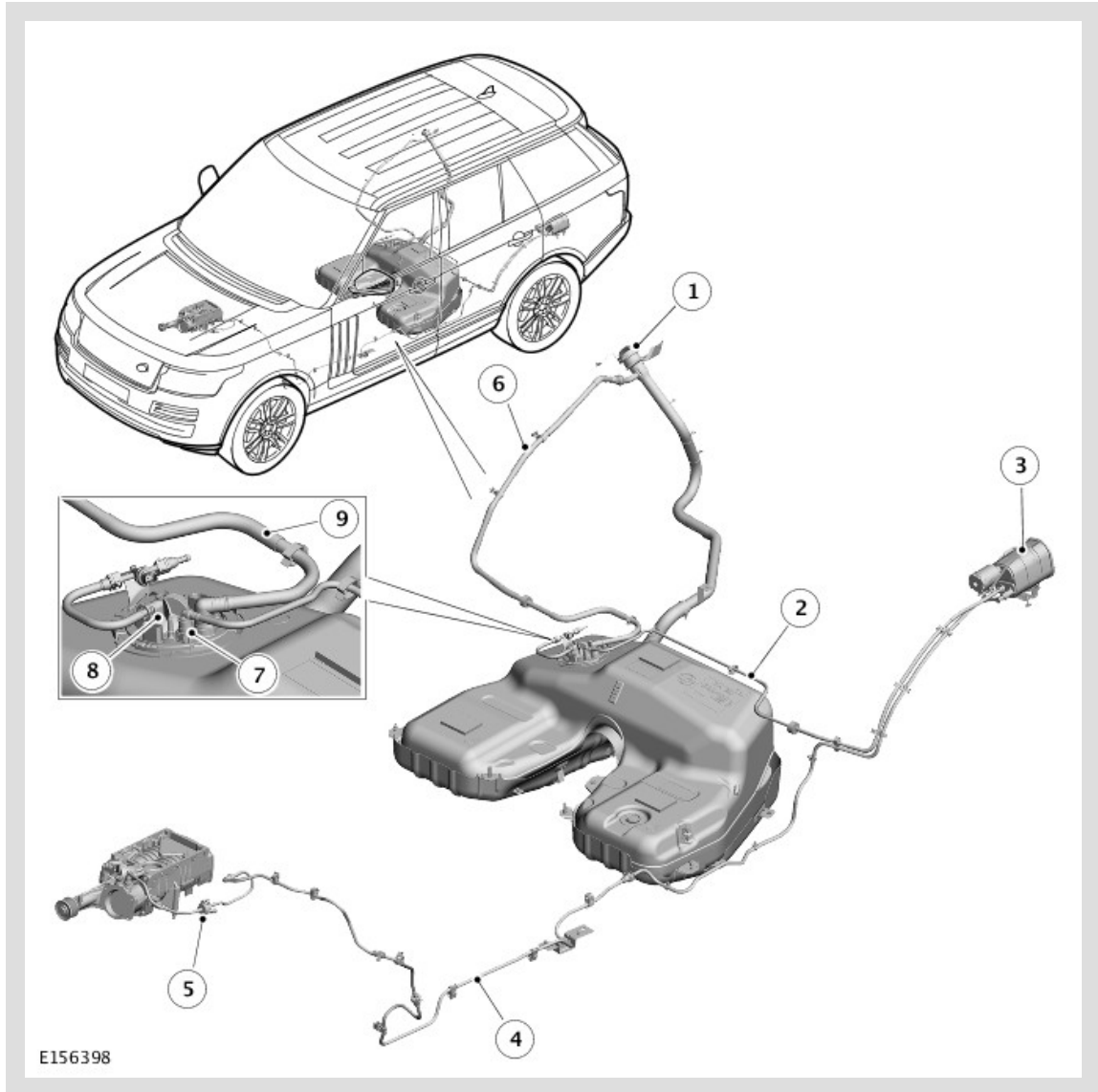
DESCRIPTION AND OPERATION

COMPONENT LOCATION

COMPONENT LOCATION - 1 OF 2 - ALL MARKETS, EXCEPT NAS

NOTE:

Installation on naturally aspirated petrol engine shown, Installation on supercharged petrol engine similar.



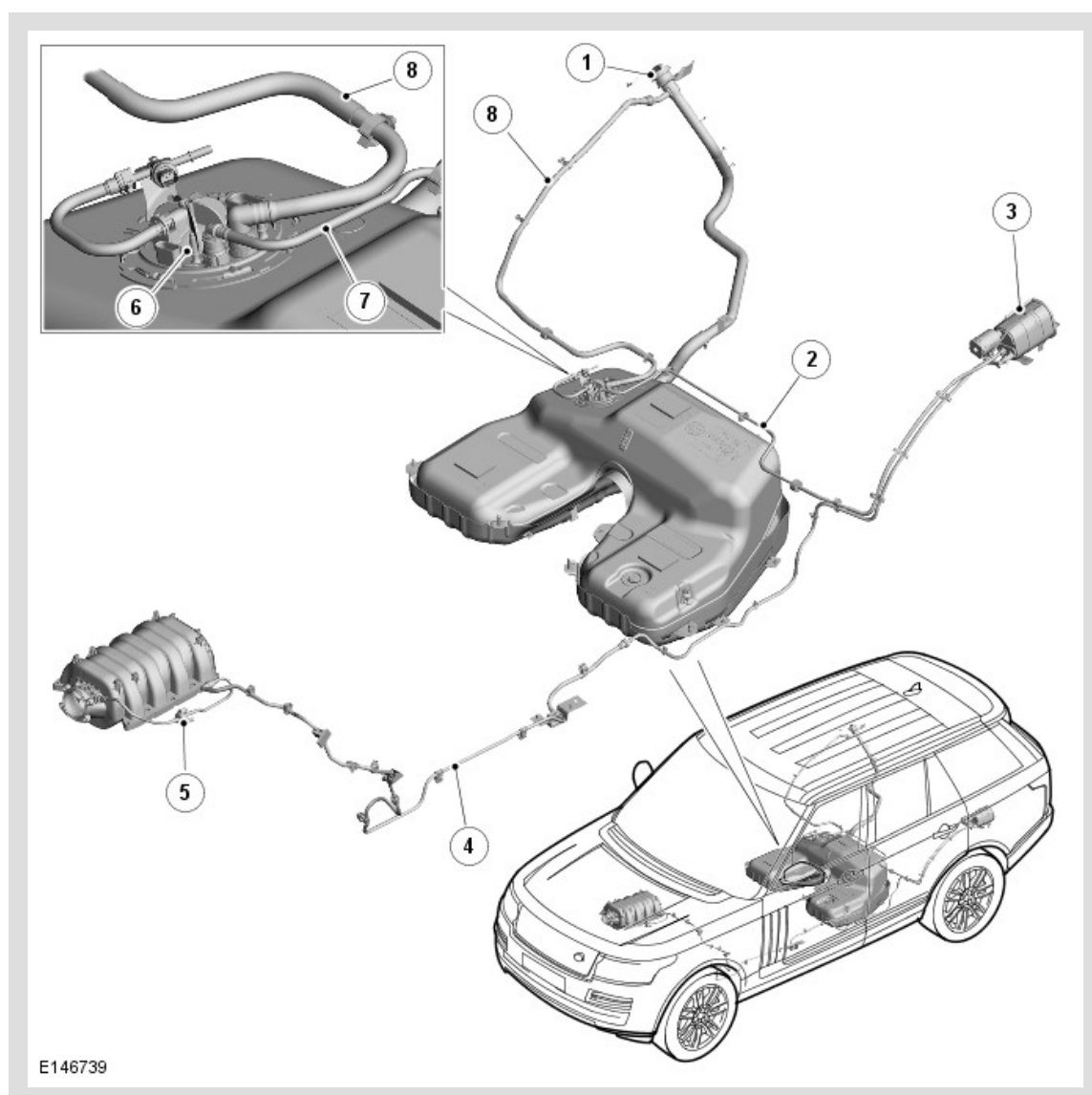
ITEM	DESCRIPTION
1	Fuel filler cap and pipe
2	Fuel tank vent pipe to evaporative emissions canister
3	Evaporative emissions canister
4	Underfloor purge line - evaporative emissions canister to purge valve
5	Purge valve with pipes
6	Refueling breather from fuel tank stub to upper filler pipe connection

7	Ventilation pipe from tank flange to evaporative emissions canister
8	Fuel pump module
9	Refueling breather from fuel tank stub to upper filler pipe connection

COMPONENT LOCATION - 2 OF 2 - NAS MARKET VEHICLES ONLY

NOTE:

Installation on supercharged petrol engine shown, Installation on naturally aspirated petrol engine similar.



ITEM	DESCRIPTION
1	Communication tube - tank to filler pipe Fuel pump module
2	Fuel filler cap and pipe
3	Refueling breather from tank stub to on-board refueling vapor recovery and evaporative emissions canister
4	Evaporative emissions canister; secondary evaporative emissions canister and Diagnostic Module Tank Leakage (DMTL) pump
5	Underfloor purge line - evaporative emissions canister to purge valve
6	Purge valve with pipes
7	Fuel pump module
8	Ventilation pipe from tank flange to evaporative emissions canister
9	Refueling breather from tank stub to on-board refueling vapor recovery and evaporative emissions canister
10	Communication tube - tank to filler pipe Fuel pump module

OVERVIEW

The Evaporative Emission (EVAP) control system reduces the level of hydrocarbons released into the atmosphere by fuel vapor venting from the fuel tank. The system comprises an evaporative emissions canister, purge valve and interconnecting vent pipes and hoses. The vent pipes are connected to the system components using quick release connectors.

On North American Specification (NAS) market vehicles, the system is checked for leak tightness by a Diagnostic Module Tank Leakage (DMTL) pump (part of the evaporative emissions canisters), which pressurizes the system and checks for leakage.

Fuel vapor is generated by the fuel in the tank and the amount of vapor produced increases as the fuel heats up. Fuel vapor can flow freely to the evaporative emissions canisters via the tank venting system. The venting system consists of a combined vent and rollover valve installed in the fuel tank. On all vehicles except NAS, the refueling breather pipes connect from the tank breather stub to the top of the fuel filler pipe, and the fuel tank vent system to the evaporative emissions canister. On NAS market vehicles, the fuel tank vent system connects into a refueling breather pipe that connects to the evaporative emissions canisters. NAS vehicles also have a communication line that connects the upper part of the filler pipe to the tank vent system.

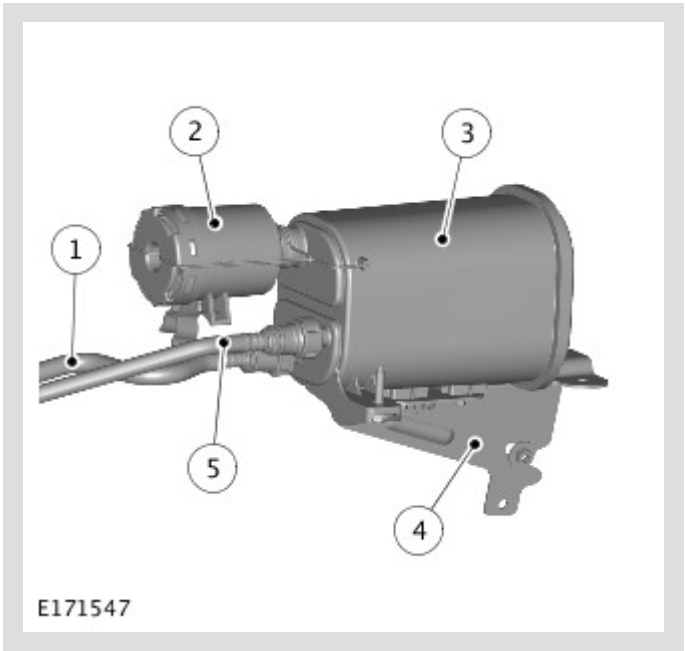
The vapor from the fuel tank passes into the evaporative emissions canisters where it is absorbed and stored by the charcoal. Because there is a limit to the amount of vapor the canisters can contain, the fuel vapor is purged from the canisters when the engine is running and burned in the engine during the combustion cycle.

On NAS market vehicles the vapor generated in the fuel tank during refueling flows without restriction to the evaporative emissions canisters. On all vehicles except NAS, the vapor is restricted in its path to the evaporative emissions canister but can flow freely during the fuel filling operation to atmosphere, via the filler opening.

DESCRIPTION

EVAPORATIVE EMISSIONS CANISTER

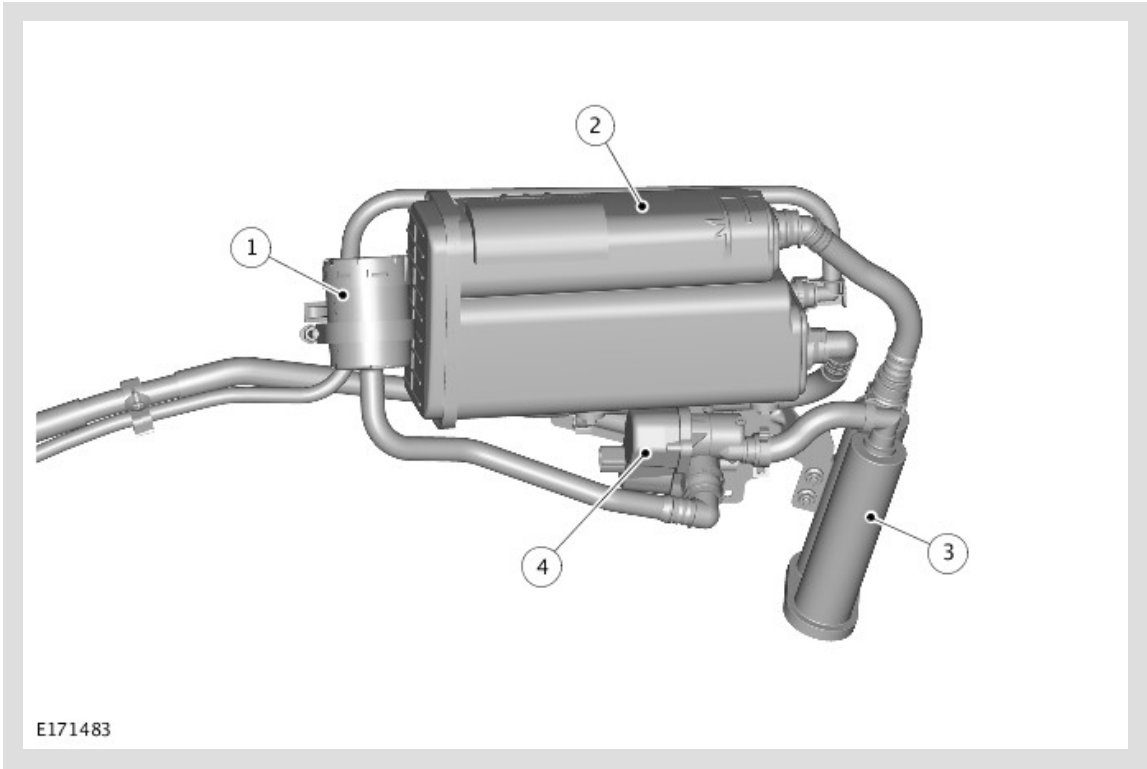
Evaporative Emissions Canister - All Markets, Except NAS



ITEM	DESCRIPTION
1	Pipe to purge valve
2	Air filter

3	Evaporative emissions canister
4	Mounting bracket
5	Pipe from fuel tank vent system

Evaporative Emissions Canister - NAS Market Vehicles Only



ITEM	DESCRIPTION
1	Air filter
2	Evaporative emissions canister
3	Secondary evaporative emissions canister
4	Diagnostic Module Tank Leakage (DMTL) pump

Evaporative emissions canister is located in the left rear wheel arch, on a bracket attached to the wheel housing inner panel. above the wheel arch splash shield. The

canister has 3 ports which allow for the attachment of an air intake filter, the pipe to the purge valve and the pipe from the tank vent. On North American Specification (NAS) vehicles the atmospheric hose connection is connected directly to the Diagnostic Monitoring Tank Leakage (DMTL) air intake filter.

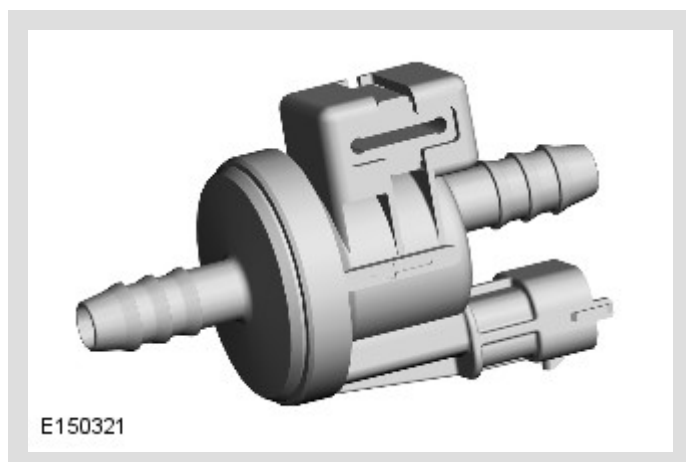
DIAGNOSTIC MODULE TANK LEAKAGE - NAS MARKET ONLY

The DMTL system periodically checks the EVAP system and the fuel tank for leaks when the ignition is switched off. The DMTL system comprises the previously described components of the Evaporative Emission (EVAP) system and a DMTL pump.

The DMTL pump incorporates an electric air pump, a Positive Temperature Coefficient (PTC) heating element, a normally open changeover valve and a reference orifice. The DMTL pump is only operated when the ignition is switched off and is controlled by the Powertrain Control Module (PCM). The PCM also monitors operation of the air pump and the changeover valve for faults.

A connection with the PCM relay in the Engine Junction Box (EJB) supplies battery power to the DMTL pump. The PCM applies separate ground connections to operate the heater element, the air pump and the changeover valve individually, as required.

PURGE VALVE



The purge valve is installed on a bracket attached to the left camshaft cover. The pipe to the engine from the purge valve is connected to the intake manifold (naturally aspirated vehicles), or supercharger front cover (supercharged vehicles), with a quick release connector. The pipe to the evaporative emission canister from the purge valve is installed between the left camshaft cover and ignition coil cover then goes down the rear of the left front suspension housing and along the underside of the vehicle and into the left rear wheel arch.

The purge valve is a solenoid operated valve, which is closed when de-energized. The valve is controlled by the PCM and is operated when engine operating conditions are suitable for purging of the evaporative emission canister.

The purge valve is controlled by a Pulse Width Modulation (PWM) signal at 10 Hz from the PCM. At this frequency, the pulses of purge gas flow into the engine in an almost

continuous flow. The valve operates between 11% and 99% duty or mark space ratio (% open time).

Atmospheric pressure is higher than the pressure at the inlet of the intake supercharger under all throttle settings and engine running conditions. It is this pressure differential that causes air to flow through the EVAP system to the engine.

OPERATION

PURGE VALVE

The Powertrain Control Module (PCM) waits until the engine is running above 40°C (104°F) coolant temperature, with closed loop fueling system, before the purging process is activated. Under these conditions the engine should be running smoothly with no warm up enrichment. The purge valve duty (and flow) is initially ramped slowly because the vapor concentration is unknown and a sudden increase in purge could cause the engine to stall or loss of air fuel ratio control to occur. The concentration is then determined from the amount of adjustment that the closed loop fueling system is required to make to achieve the target air fuel ratio. Once the concentration has been determined, the purge flow can be increased and the injected fuel can be proactively adjusted to compensate for the known purge vapor and the target air fuel ratio control is maintained.

DIAGNOSTIC MODULE TANK LEAKAGE - NAS MARKET ONLY

To check the fuel tank and the EVAP system for leaks, the PCM operates the DMTL air pump and monitors the current draw. Initially, the PCM establishes a reference current by pumping air through the reference orifice and back to atmosphere. Once the reference current is determined, the PCM closes the changeover valve, which seals the EVAP system. The purge valve remains de-energized and is therefore closed. The output from the air pump is diverted from the reference orifice and into the EVAP system.

Providing there are no leaks, the air pump will begin to pressurize the EVAP system and the load and current draw of the air pump motor increases. By monitoring the rate and level of the current increase, the PCM can determine if there is a leak in the EVAP system.

During normal vehicle operation, 15 seconds after the engine has started, the PCM energizes the heating element in the pump to prevent condensation formation and possible incorrect readings. The heater remains energized until either the engine or ignition are off (if no DMTL test is running) or until after the DMTL test is completed.

Leaks are classified as:

- Minor - equivalent to a hole diameter of 0.5 to 1.0 mm (0.02 to 0.04 in)

- Major - equivalent to a hole diameter of 1.0 mm (0.04 in) or greater.

The PCM performs a check for major leaks each time the ignition is switched off, providing the following conditions are met:

- The vehicle speed is zero.
- The engine speed is zero.
- The pressure altitude (70 kPa (10.15 lbf/in²) derived from engine load calculations) is below 3047 m (10000 feet).
- The ambient temperature is between 0 and 40°C (32 and 104°F).
- The evaporative emission canister vapor concentration factor is less than 3 (where 0 is no fuel vapor, 1 is stoichiometric fuel vapor and greater than 1 is rich fuel vapor).
- The fuel tank level is valid and between 15 and 85% of nominal capacity.
- The battery voltage is between 11 and 16 V.
- The last engine running time was more than 10 minutes (to ensure any moisture within the DMTL pump has been removed).
- During the last engine start the difference between engine coolant temperature and ambient air temperature was less than 7°C (13°F).
- No errors are detected with the EVAP system components, the ambient air temperature and the fuel level.
- High range is selected on the transfer case.

 NOTE:

A leak test can be performed using the Land Rover approved diagnostic equipment. This overrides the above conditions and is useful for checking correct system and component operation.

The PCM performs a check for minor leaks after every 2nd major leak check.

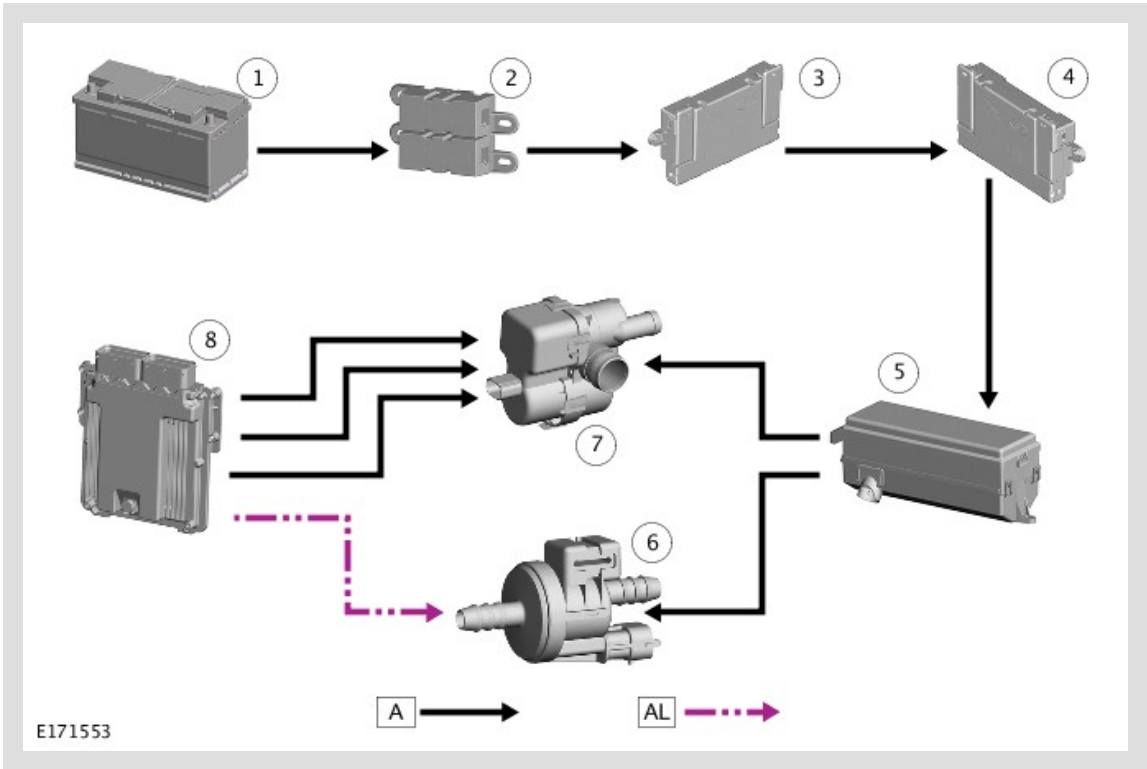
When the leak check is complete, the PCM stops the DMTL pump and opens (de-energizes) the changeover valve.

If the fuel filler cap is opened or refueling is detected during the leak check, by a sudden drop in the current draw or a rise in the fuel level, the PCM aborts the leak check.

If a leak is detected during a test, the PCM stores an appropriate Diagnostic Trouble Code (DTC) in its memory. If a leak is detected on two consecutive tests, the PCM illuminates the Malfunction Indicator Lamp (MIL) in the Instrument Cluster (IC) on the next drive cycle.

The duration of a leak check can be between 60 and 900 seconds depending on the test results (developed tank pressure amperage within a specific time period) and fuel tank level.

CONTROL DIAGRAM



A = HARDWIRED; AL = PULSE WIDTH MODULATION (PWM).

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Auxiliary Junction Box (AJB)
5	Engine Junction Box (EJB)
6	Purge valve
7	Diagnostic Module Tank Leakage (DMTL) pump (NAS only)
8	Powertrain Control Module (PCM)

